

Ganesh Sharma

Software Engineer (Python / FastAPI / ReactJS / PostgreSQL)

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GitHub | Portfolio | LinkedIn

PROFILE SUMMARY

Software Engineer focused on complete end-to-end full-stack development, REST APIs, databases, and AI applications. Hands-on experience with FastAPI, PostgreSQL, SQLAlchemy, ReactJS, JavaScript, and Tailwind CSS, with working knowledge of Docker, Git, AWS, and testing. Focused on building clean, maintainable, scalable products and reliable software solutions.

PROJECTS

CampusSphere: Multi-user Student & Academic Management System | *FastAPI, ReactJS, PostgreSQL, SQLAlchemy, Pydantic, Alembic, Pytest, Docker*

- Built a centralized academic management platform to handle students, teachers, departments, attendance, fees, and administrative operations while preventing unauthorized users from accessing or modifying sensitive institutional data.
- Designed and developed the system end-to-end with FastAPI, ReactJS, PostgreSQL, and SQLAlchemy, implementing JWT-based authentication, protected endpoints, role-based access control (RBAC), strict Pydantic request/response validation, relational constraints, transaction-aware database operations, optimized ORM queries to avoid $N + 1$ patterns, reliable Alembic migrations, comprehensive Pytest coverage, and Docker-based containerization.
- Delivered a production-oriented, multi-user system with clearly separated responsibilities like administrators manage institutional data and staff, teachers manage attendance and academic operations, and clerks/accountants manage fees—providing a secure and maintainable foundation for real-world academic workflows.

Medium-like Blog Writing Platform | *FastAPI, ReactJS, PostgreSQL, SQLAlchemy, Pydantic, Alembic, Tailwind CSS*

- Built a Medium-inspired publishing platform that enables writers to create and manage long-form content while allowing readers to discover and read published blogs through a centralized web application.
- Developed the platform end-to-end with FastAPI, ReactJS, PostgreSQL, and SQLAlchemy, implementing authenticated writer workflows for creating, editing, and managing blogs, protected API endpoints, Pydantic request/response validation, relational database models, and Alembic migrations for reliable schema evolution.
- Delivered a full-stack content platform with secure user access, writer-owned content management, persistent PostgreSQL storage, and a responsive React/Tailwind interface, providing a complete publishing workflow from content creation to reader consumption.

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS

Backend Development: FastAPI, Pydantic, SQLAlchemy, Alembic, REST APIs, Database Design

Frontend Development: ReactJS, Tailwind CSS, Responsive Design, Figma

Databases: PostgreSQL, SQLite, Redis

DevOps, Cloud, & Testing: Git/GitHub, Docker, Linux Commands, Nginx, AWS (EC2, S3, RDS, S3), Pytest

AI/ML: LLM APIs, LangChain

RESEARCH & PUBLICATIONS

MITRA AI Chatbot Assistant: Using 8-Bit Quantized LLMs on Consumer-Grade GPUs

Published in *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*

Volume 14, Issue III, pp. 5059–5065, ISSN: 2321-9653.

EDUCATION

University of Mumbai

B.E. in Computer Science & Engineering (AI & ML)

Mumbai, India

2022–2026